SafeCart: AI-Powered Allergen Detection for Safe Grocery Shopping

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The Challenge: Food Allergies Affect Millions

The Global Burden

Over 200 million people worldwide manage food allergies, spending 30-60 minutes per shopping trip reading complex ingredient labels. Hidden allergens under technical names like "sodium caseinate" (dairy) and "triticum" (gluten) make label reading hazardous and exhausting.

The Core Problems

- Dense, complex packaging with varying fonts and layouts
- Technical ingredient names unrecognized by consumers
- Time pressure leading to reading errors
- Accessibility barriers for visually impaired individuals

Introducing SafeCart: Your AI Shopping Assistant

SafeCart combines Optical Character Recognition (OCR) and Natural Language Processing (NLP) to automatically identify potential allergens in packaged food products. Users simply upload product images, and the system extracts ingredient text, detects allergens, and provides personalized safety classifications with confidence scoring.

Image Capture

Upload product packaging photos

OCR Processing

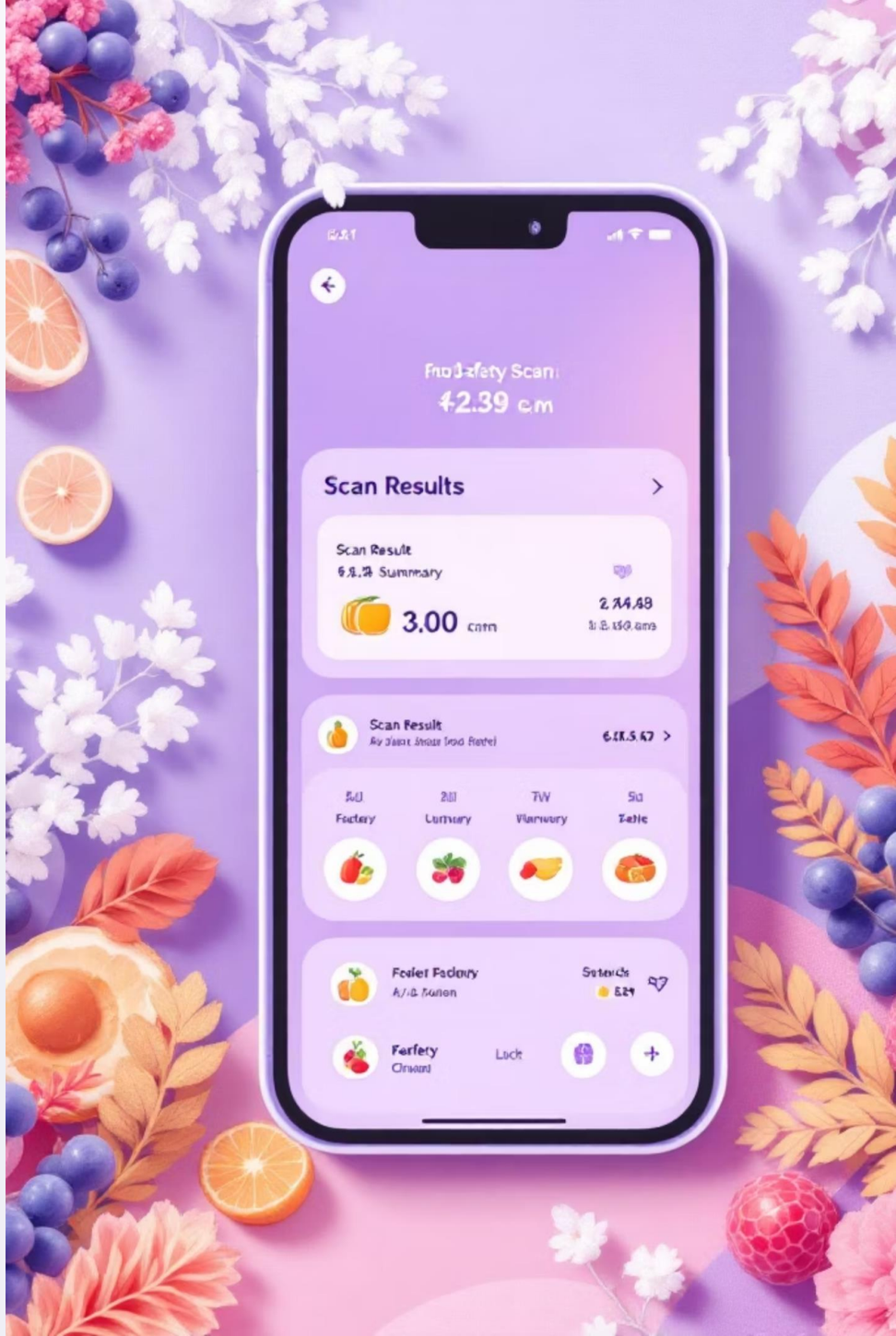
Extract ingredient text automatically

NLP Analysis

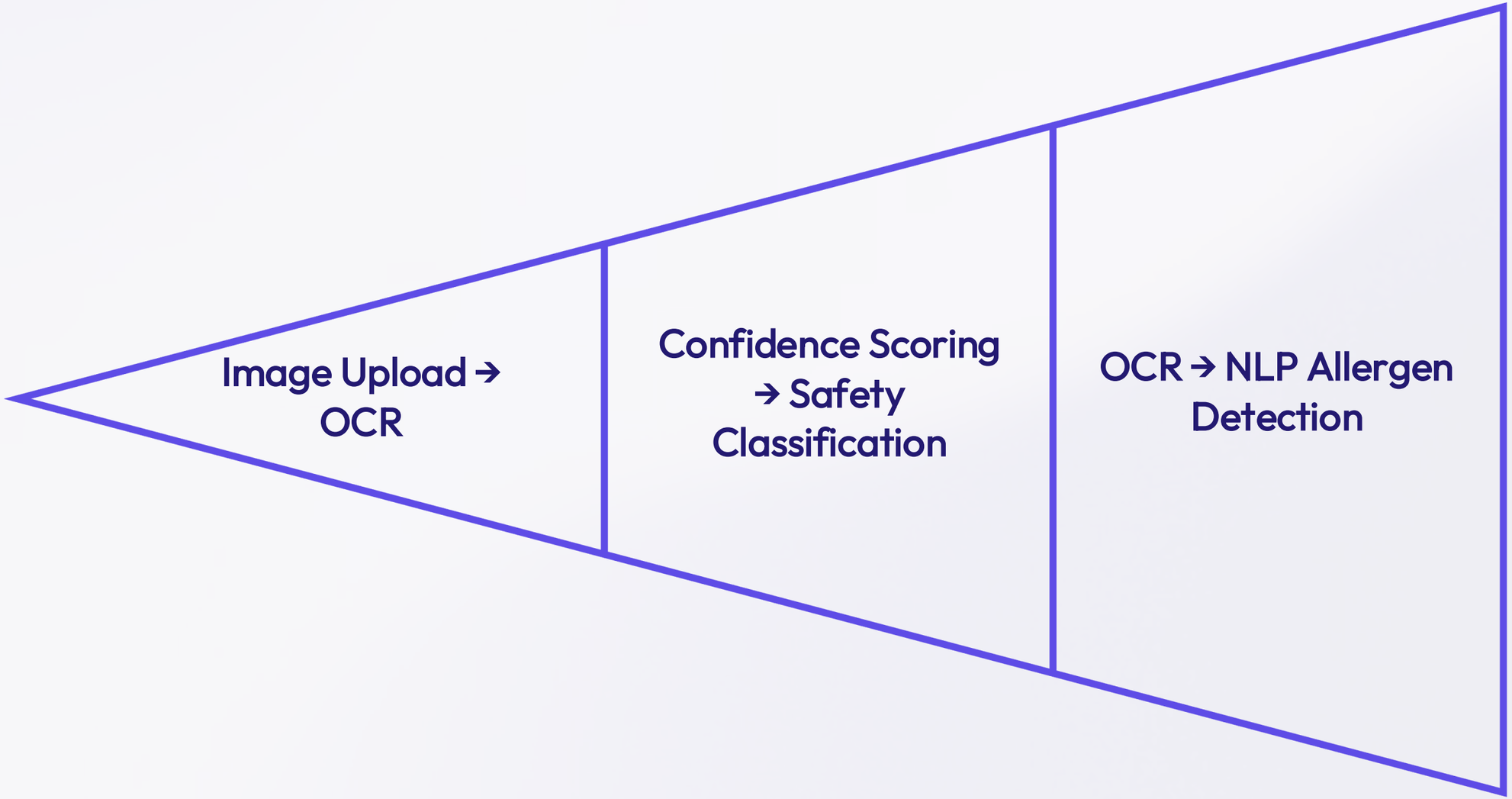
Identify allergen keywords and synonyms

Safety Rating

Personalized safe/unsafe classification



System Architecture: Three-Stage Pipeline



Key Features and Capabilities

Comprehensive Allergen Database

Supports milk, gluten, soy, peanuts, tree nuts, eggs, fish, shellfish, and sesame with extensive synonym mappings for technical ingredient names.

Personalized User Profiles

Custom allergen filtering based on individual user profiles with confidence scoring for detection reliability assessment.

Visual Safety Indicators

Intuitive Safe/Unsafe/Unknown classifications with color coding and extracted text with detected allergens clearly highlighted.

Technology Stack

1

Backend Framework

FastAPI (Python) for high-performance web server with automatic API documentation

2

OCR Engine

EasyOCR for multilingual text extraction with GPU acceleration capability

3

Image Processing

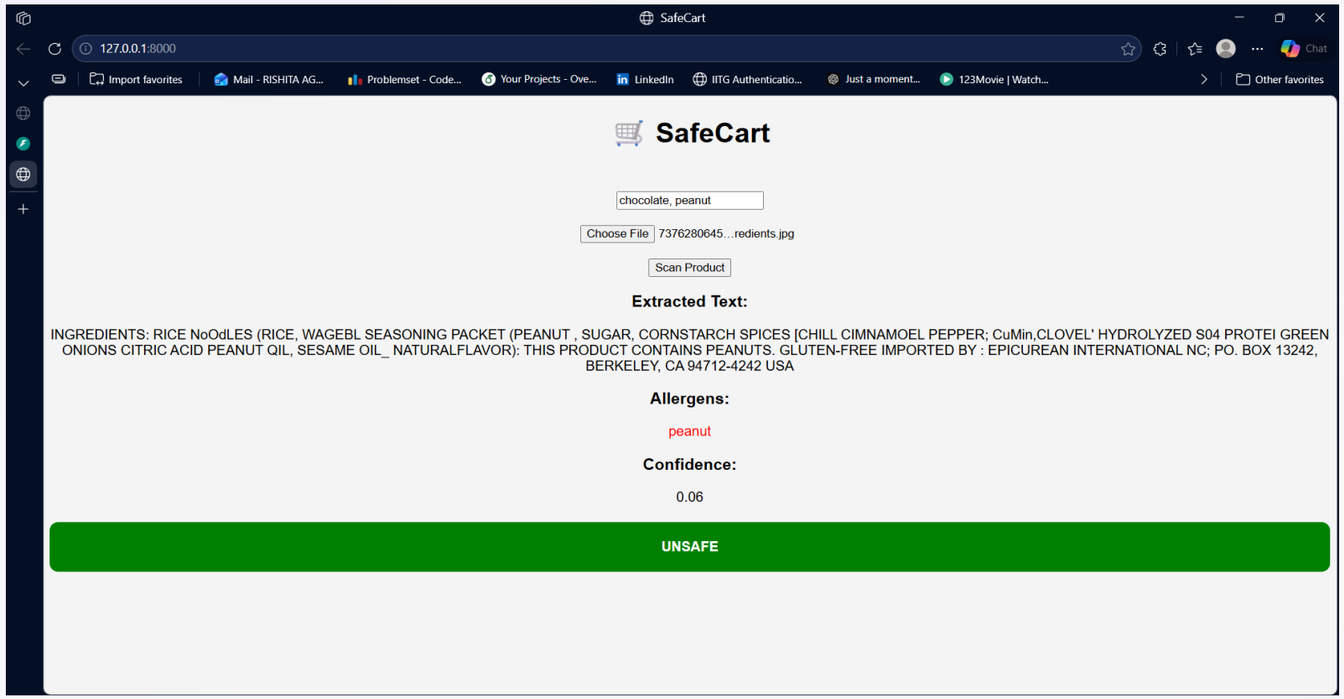
OpenCV for preprocessing including noise reduction and contrast enhancement

4

Frontend Interface

HTML, CSS, JavaScript for responsive web application with real-time results display

Experimental Results



Performance Summary

Evaluated on 50 product images across various categories including snacks, dairy, beverages, and bakery products under different lighting conditions.

63.13%

OCR Accuracy

Moderate performance impacted by variable image quality and complex packaging designs

83.33%

Precision

High precision indicates most detected allergens were correct identifications

35.71%

Recall

Limited recall due to OCR errors and incomplete text extraction

50%

F1-Score

Balanced metric reflecting reasonable precision and limited recall

0.34s

Processing Time

Excellent computational efficiency enabling real-time interactive use

Challenges and Limitations

OCR Dependency

System accuracy fundamentally limited by OCR performance, which degrades with poor image quality, blurred photographs, and reflective packaging materials causing glare.

Image Requirements

Product images must capture ingredient lists; front-facing images without ingredient sections yield incomplete or no analysis. Complex packaging with curved text challenges extraction.

Rule-Based NLP

Pattern-matching approach may miss complex or uncommon ingredient formulations that trained machine learning models could capture through contextual understanding.

Language Support

Current implementation optimized for English text with limited multilingual support, restricting usability for international products and diverse populations.

Future Enhancements

Region Detection

Implement YOLO object detection to automatically locate ingredient label regions before OCR processing

Multilingual NLP

Integrate XLM-RoBERTa for ingredient parsing across multiple languages

Cloud OCR

Integrate Google Vision API for improved accuracy on challenging images

Mobile Applications

Develop native iOS and Android apps with camera integration for real-time scanning and augmented reality overlays displaying safety information on live camera feeds.

Database Integration

Connect to OpenFoodFacts for supplementary product information, cross-validation, and safe grocery list generation recommending allergen-free alternatives.

Impact and Conclusion

Societal Impact

SafeCart addresses a genuine humanitarian need, potentially reducing accidental allergen exposure incidents, improving food autonomy for elderly and disabled populations, and assisting international travelers navigating unfamiliar food products.

Academic Relevance

Demonstrates practical multimodal AI integration combining computer vision and natural language processing for real-time inference pipeline design and user-centric AI system development.

> "SafeCart embodies the core promise of multimodal AI: combining heterogeneous sensory and linguistic inputs to solve real-world problems that improve human health and safety."

Key Achievements: End-to-end pipeline achieving 83.33% precision and 50% F1-score on allergen detection, real-time processing with 0.34-second latency, intuitive personalized user interface, and extensible architecture supporting future enhancements through deep learning, multilingual support, and mobile deployment.

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Thank you

Any Questions?